Microbial Ecology (2023) 85:317–329
https://doi.org/10.1007/s00248-021-01951-0

HUMAN MICROBIOME

Influence of Environmental Factors on Salivary Microbiota and Their Metabolic Pathway: Next-Generation Sequencing Approach

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Received: 21 October 2021 / Accepted: 16 December 2021 / Published online: 11 January 2022
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Abstract

The current study aimed to investigate the effect of periodontitis and long-term heavy metal (HM) exposure on the salivary microbiome. The patients were divided into four groups as Wu Wei control (WWC) group involved healthy individuals, Wu Wei periodontitis (WWP) patients having periodontitis, Jing Chang with metal pollution periodontally healthy individuals (JCP), and Kuang periodontitis (KP). The most abundant bacteria identified at the phylum level in the WWC group were Bacteroides, Firmicutes, and Fusobacteria. Firmicutes were observed in a significantly higher proportion in the KP group than in the WWC, WWP, and JCP. At the genus level, the WWC has major dominating bacterial genera (such as *Leptotrichia*, *Neisseria*, and *Fusobacterium*) which were similar to WWP and KP group. The significant difference ($p < 0.05$) was found in alpha diversity while in beta diversity, the significant ($p = 0.005$) results were found among the four groups. The correlation of oral microbiota revealed that HMs present in the soil (Cr, Ni, and Cu) are associated with the growth of *Capnocytophaga*, *Selenomonas*, *Aggregatibacter*, and *Campylobacter*. The bacterial functions in the KP group were higher in translation and nucleotide metabolism than in the WWP group. This demonstrated that long-term exposure to HMs can influence the salivary microbiota which can alter the functioning, and diversity of bacteria.

Keywords Heavy metal · Periodontitis · Bacterial taxa · Saliva · Microbiota · Firmicutes

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Introduction

The pollution of heavy metals (HMs) is a serious threat to the environment and has become one of the major concerns due to the adverse effects on human and environmental health worldwide [1]. These inorganic pollutants accumulated and discharged in soil and water bodies can enter into the atmosphere ultimately in the human body through the food chain, oral ingestion, and direct contact with skin [2]. A high concentration of mercury (Hg) can damage the human nervous system [3]. An excessive amount of cadmium (Cd) can impair kidney function [4], while nickel can cause an allergic reaction, even tumorigenesis [5].

Periodontitis is the most common inflammatory disease initiated by the oral microbiome, which can progressively lead to the periodontal tissues, damaging and resulting in the formation of a periodontal pocket [6]. Periodontitis is a chronic infective disease caused by ecological disturbances of plaque microbes that triggers an inflammatory response, and the inflammatory cytokines, in turn, alter the microbial ecology. Specifically, periodontitis is caused by the effect

of the bacterial structure and metabolic products, which are involved in the inflammatory responses of host cells, on the normal oral functions. Its mechanism of disease is still unclear and a few studies have indicated its association with microorganisms, the environment, and other aspects. In the field of oral ecology, research on periodontitis associated with the aforementioned factors has become the hotspot [7]. The previous study revealed that a high content of lead (Pb) and Hg in blood cause loss of attachment in periodontal tissues and ultimately increase the potential risk factors for periodontitis [8].

The recent studies have reported that the cigarette smoke consumption (active or passive), which contains several toxic metals (such as Ni, Cd, Cr, Mn, and Pb), can damage to periodontal tissues and oral epithelium [9]. These impacts can cause serious threat to human health which includes enhancement of inflammatory mediator levels, and lower down the repair capacity of periodontal ligament cells [10]. These findings provide baseline information for the correlation between HM exposure and periodontitis. Therefore, it is necessary to understand the biological mechanisms of the development of periodontal disease due to HM exposure.

The Jinchang City, located in Gansu Province, China, is known as the "nickel capital" of China. Due to industrialization in a particular region, several HMs are released into the environment, which poses a serious threat to human health [11]. The current study is focused on the relationship between oral microbiota and HM accumulation. Initially, the saliva samples were collected from patients belonging to different regions (including the Jinchang mining area, Jinchang City, and Wuwei area). Which is followed by analyzing the salivary microbiota structure through Illumina MiSeq. The soil samples were collected from the same area where the patients abide. The HM concentration was assessed along with the impact of HM exposure on the bacterial community.

Materials and Methods

Patient Selection

The participants in the present study were chosen from the patients who underwent a physical examination. The patients chosen for current study were selected from the Oral Department of Wuwei People's Hospital, Jinchang Baye Hospital, and people working in Jinchang nickel smeltery. All the participants were aged between 32 and 56 years. The patients with systematic diseases, antibiotics prescribed from last 3 months, alcoholics, periodontal treatment history in the last 6 months' dental caries, and pregnant women's or with breastfeeding period were excluded. The selected periodontitis criteria were patients whose mouth had at least 4 teeth with 1 or multiple sites of PD ≥ 4 mm, CAL ≥ 1 mm, and probing bleeding were diagnosed as having periodontitis. In total, 34 individuals were selected and divided into four groups on the basis of the presence of periodontitis and exposure to HMs: (1) WWC (Wu Wei control) group ($n=9$), which was considered the normal control group, comprised periodontally healthy individuals residing in Wuwei District with no HM pollution; (2) WWP (Wu Wei periodontitis) group ($n=9$) comprised the patients diagnosed as having periodontitis and residing in Wuwei District; (3) JCP (Jing Chang with metal pollution periodontally) group ($n=7$) comprised periodontally healthy individuals from the Jinchang mining area; and (4) KP (Kuang periodontitis) group ($n=9$) comprised the patients diagnosed as having periodontitis and residing in the Jinchang mining area. This study was approved by the Medical Ethics Committee of the Lanzhou University Second Hospital (2018A-100), and all the participants provided written informed consent.

Saliva Sample Collection

Saliva sample (4 mL) was collected from each individual. Before sample collection, the participants were restrained from drinking or eating and were asked to wash their mouth with drinking water 30 min. After sample collection, the saliva sample was preserved at − 80 ℃ for further investigation.

Collection of Soil Sample and Heavy Metal Analysis

A total of 20 representative sampling sites were selected, and soil samples with equal mass were collected from 0 to 20 cm under the surface at each site. One kilogram of soil samples was collected in the plastic bags and transfer immediately to the portable − 80° refrigerator for making the sample more convenient to store. Followed by, the samples were transferred to the laboratory under same conditions and the plant residues and stones were removed. The samples were then air-dried to form fine powder for heavy metal detection. The concentration of HMs (including Ni, Cr, and Cu) was determined using an atomic absorption spectrophotometer (SOLAAR M6, Thermo Electron Corporation) [5]. All the experiments were conducted in the Key Laboratory of Western China's Environmental Systems, Ministry of Education, Lanzhou University, China.

DNA Extraction, Library Construction, and Sequencing

The 300-μL saliva sample was used for bacterial DNA extraction. The DNA was isolated from saliva samples by using MOBIO DNA isolation kit and second-generation MN Kit, according to the microbial full-length DNA extraction protocol. The 16S V3–V4 region primers were

used as forward 338F 5′ACTCCTACGGGAGGCAGC A-3′ and reverse primer 806R ′5GGACTACHVGGGTWT CTAAT-3′ [12]. The construction of sequencing libraries and paired-end sequencing was performed on an Illumina MiSeq platform at Biomarker Technologies Co, Ltd. (Beijing, China) according to standard protocols. The primers, designed based on conservative fragments, were ligated with sequencing adaptors and used for polymerase chain reaction amplification. The amplicons were purified, quantified, and homogenized to prepare the sequencing library, and the qualified amplicons were sequenced using the Illumina MiSeq 2500 platform [13].

Data Preprocessing

The data preprocessing was conducted on the BMK CLoud platform. Paired-end (PE) sequences obtained from HiSeq sequencing were merged into a single sequence having one tag according to the overlapping relationship between PE reads. Thereafter, the reads and the merging effect were quality-controlled and filtered. The whole process was completed in three steps: (1) splicing of PE reads: raw tags were obtained by using the FLASH software (version 1.2.7) to splice the reads of each sample through the overlapping relationship; (2) filtering of tags: clean tags were obtained by using the Trimmomatic software (version 0.33) to filter the spliced raw tags; and (3) removal of chimerism: the final effective tags were obtained by using the UCHIME software (version 4.2) to identify and remove chimeric sequences.

Statistical Analysis

Analysis of variance (ANOVA) was used to evaluate the relative bacterial abundance, and the Wilcoxon test was used to compare the alpha diversity index between the four groups. The Beta diversity was analyzed using the principal coordinate analysis (PCoA) based on weighted and unweighted algorithms at the operational taxonomic unit (OTU) level. The diversity between the four groups was evaluated using permutational multivariate analysis of variance (PERMANOVA). The correlation between microbiota (microbial community in saliva) and environmental factors (including HM pollution and pH) of the soil samples was analyzed. The threshold value of the correlation coefficient was set at 0.1 at the genus level. The differences between the groups were assessed using the linear discriminant analysis effect size (LEfSe). An LDA score of >2 was considered as the cut-off value for biomarker screening. At the genus level, a threshold p-value of <0.05 and SparCC were used for constructing a correlation plot. The picrust2 was used for Kyoto Encyclopedia of Genes and Genomes (KEGG) function prediction, and SPSS (version 22.0) was used for all statistical analyses.

Results and Discussion

Bacterial Community Structure of Saliva Samples

High-throughput sequencing was performed for KP, JCP, WWP, and WWC groups to monitor the changes in the salivary microbial community. A total of 12 phyla were observed at phylum level in which the dominant bacterial populations (%) observed in the four groups (KP, JCP, WWP, and WWC) were Bacteroidetes (24.03%, 24.58%, 23.44%, and 25.95%), Firmicutes (17.01%, 15.99%, 25.78%, and 27.54%), Fusobacteria (19.89%, 19.60%, 19.15%, and 25.06%), Proteobacteria (23.95%, 26.23%, 17.38%, and 8.16%), Actinobacteria (9.30%, 9.04%, 8.41%, and 5.87%), and Patescibacteria (1.84%, 2.09%, 2.55%, and 2.75%), respectively (Fig. 1). At the phylum level, the bacterial composition of the experimental group was found to be comparable to that of the control group (WWC). The dominant or the most abundant bacteria identified in the control group were Bacteroides, Firmicutes, Fusobacteria, and Proteobacteria. In particular, Firmicutes were observed in a significantly higher proportion in the KP group than in the control groups and the other two groups. The phylum Firmicutes comprise a large number of genera, which are present in either healthy or diseased states. These findings resemble the previous studies and suggest the reliability of results [14–17]. The abundance of Fusobacteria was significantly higher in the JCP group. The proportion of the predominant bacteria Proteobacteria was found to be the highest in the WWP group, which is almost similar with control group. Spirochaetes is a periodontal pathogen; although it was not identified as the predominant bacteria, its abundance was found to be the highest in the control group which was quite less and almost equivalent to KP group. The periodontal microbiota varies across different environmental conditions [18]. Spirochaetes are prominent in the microbial infections that cause periodontitis. The differences in the bacterial abundance observed in this study may be attributed to differences in the region of residence of the study population [19]. The ANOVA results indicated the highest abundance of the phylum Firmicutes in the KP group ($p<0.001$), Proteobacteria in the WWP group ($p=0.002$), and Spirochaetes in the WWC group ($p=0.014$). All these differences in bacterial abundance were found to vary significantly from those in other 3 groups (Table 1).

The microbial community of saliva at the genus levels is described in Fig. 2. At the genus level, the control group has major dominating bacterial genera which were *Leptotrichia*, *Neisseria*, *Fusobacterium*, *Capnocytophaga*, *Aggregatibacter*, and *Corynebacterium.* The populations (%) observed in the four groups (KP, JCP, WWP, and

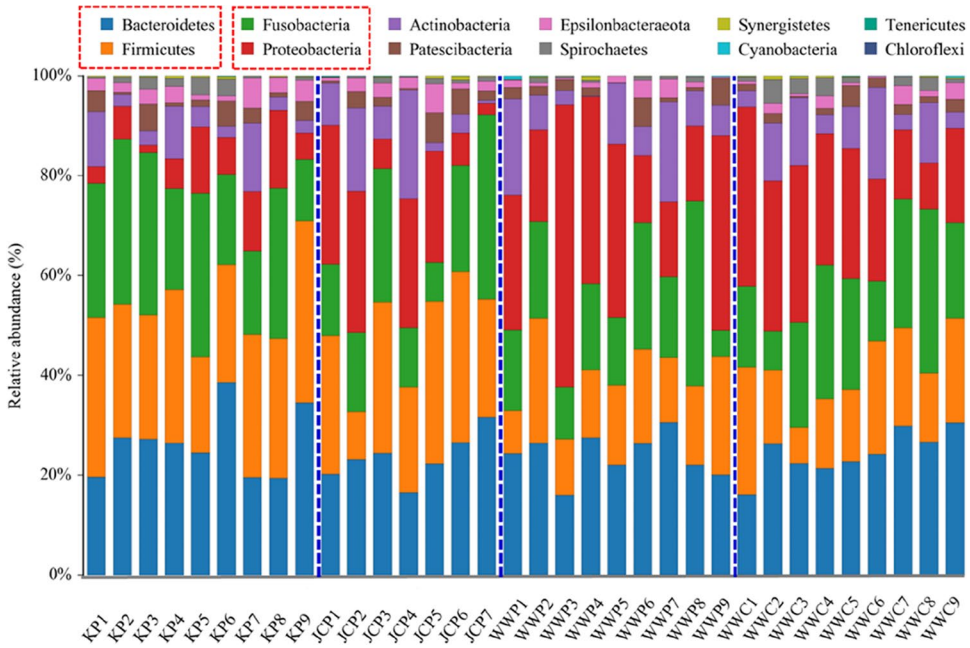


Fig. 1 Changes in bacterial community under various groups. The % of relative abundance at phylum level showed microbial taxa of all the groups

Table 1 The framework of bacterial community analysis of saliva sample along with ANOVA results indicated the highest abundance ($p < 0.05$)

Bacterial abundance	Jing Chang with metal pollution periodontally (JCP)	Kuang periodontitis (KP)	Wu Wei periodontitis (WWP)	Wu Wei control (WWC)	*p*-value
Firmicutes[a]	0.255946 ± 0.084784	0.27781 ± 0.0501810	0.161996 ± 0.055024	0.170207 ± 0.056381	0.000452
Proteobacteria[a]	0.1701 ± 0.11553600	0.079061 ± 0.047423	0.285115 ± 0.146089	0.235813 ± 0.086947	0.001637
Spirochaetes[a]	0.006518 ± 0.004449	0.015337 ± 0.012893	0.005091 ± 0.004142	0.021004 ± 0.015656	0.014439
Selenomonas[b]	0.012487 ± 0.012792	0.028256 ± 0.012001	0.005464 ± 0.006167	0.005391 ± 0.00357	0.000020
Anaeroglobus[b]	0.002848 ± 0.002978	0.014097 ± 0.008619	0.002692 ± 0.004977	0.001371 ± 0.001787	0.000057
Howardella[b]	0.000071 ± 0.000098	0.000192 ± 0.000215	0.000015 ± 0.000029	0.000004 ± 0.000007	0.008975
Prevotella[b]	0.041697 ± 0.026299	0.049636 ± 0.028130	0.017985 ± 0.015827	0.021208 ± 0.014777	0.011038
Neisseria[b]	0.067774 ± 0.076396	0.030934 ± 0.034443	0.144879 ± 0.090317	0.099689 ± 0.056143	0.008915
Treponema-2[b]	0.006519 ± 0.004449	0.015338 ± 0.012893	0.005093 ± 0.004141	0.021008 ± 0.015654	0.014415
Gemella[b]	0.002622 ± 0.001621	0.001333 ± 0.000835	0.003384 ± 0.003091	0.006245 ± 0.004234	0.007652
Selenomonas-4[b]	0.003822 ± 0.003679	0.006842 ± 0.004527	0.001737 ± 0.00203	0.002163 ± 0.001669	0.007427
Johnsonella[b]	0.004045 ± 0.001493	0.003713 ± 0.003005	0.005979 ± 0.005356	0.010836 ± 0.007667	0.024977
Solobacterium[b]	0.002347 ± 0.002358	0.000757 ± 0.000546	0.000744 ± 0.001131	0.00041 ± 0.0003870	0.024909
Peptostreptococcus[b]	0.007681 ± 0.009757	0.002911 ± 0.001731	0.001396 ± 0.002024	0.001071 ± 0.001001	0.033530
Capnocytophaga[b]	0.055337 ± 0.055249	0.030965 ± 0.024468	0.099202 ± 0.072471	0.102543 ± 0.061143	0.030993
Eikenella[b]	0.004533 ± 0.005826	0.002538 ± 0.003796	0.002718 ± 0.002244	0.008693 ± 0.006400	0.036728
Veillonella[b]	0.09936 ± 0.0708670	0.094192 ± 0.068748	0.042701 ± 0.058342	0.028146 ± 0.035123	0.041288

[a]Bacterial abundance difference at the phylum level, [b]Bacterial abundance difference at the genus level

WWC) were *Leptotrichia* (10.84%, 11.33%, 11.88%, and 14.92%, respectively), *Fusobacterium* (9.05%, 8.27%, 7.25%, and 10.14%, respectively), *Neisseria* (10.40%, 13.65%, 7.08%, and 3.28%, respectively), *Capnocytophaga* (9.91%, 9.79%, 5.88%, and 3.26%, respectively), *Veillonella* (2.73%, 3.73%, 10.48%, and 9.22%, respectively), and *Corynebacterium* (7.15%, 6.04%, 5.19%, and 3.92%, respectively) (Fig. 2). The significant dominating bacteria at genera level in the control group were mainly *Leptotrichia*, *Fusobacterium*, *Neisseria*, and *Capnocytophaga* which is almost similar with WWP group. In a few previous studies, it is reported that the genera *Leptotrichia*, *Selenemonas*, and *Eubacterium* belonging to the phylum Firmicutes are responsible for caries and periodontal

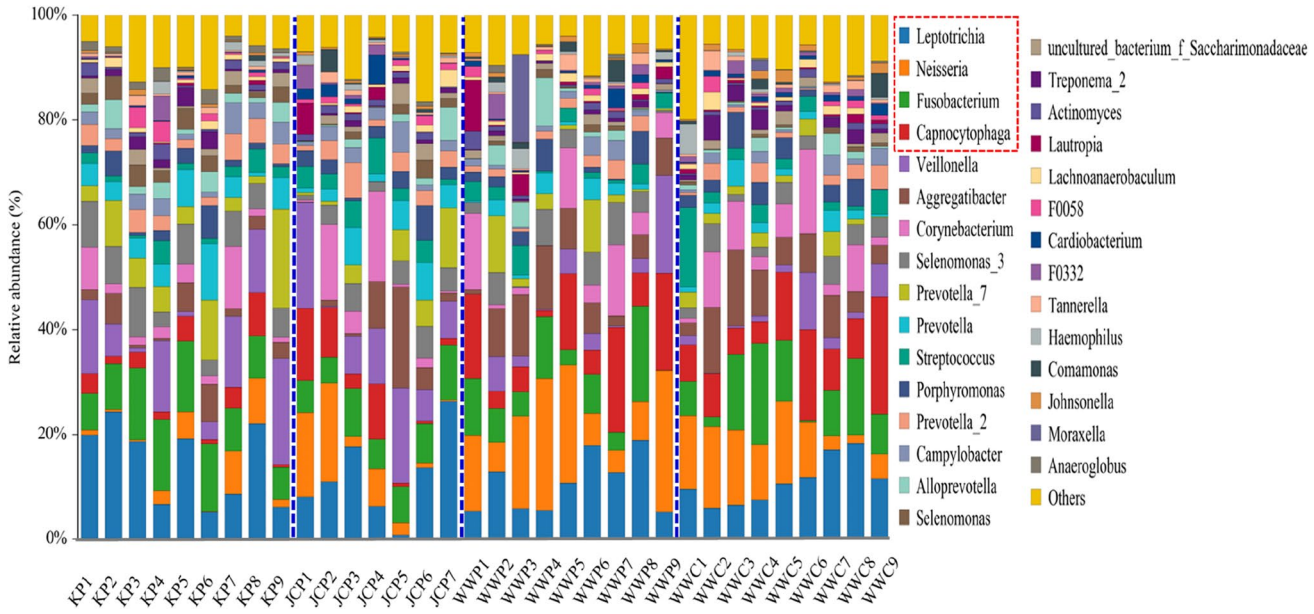


Fig. 2 Histograms show the relative abundance of the top 30 annotated microbial taxa for each sample at genus levels. The vertical and horizontal axes represent the relative abundance of microbial taxa and the sample information of four groups, respectively

disease [20–22]. *Leptotrichia* is regarded as an opportunistic pathogen prevalent in various human infections which includes periodontitis [23]. The abundance of 6 genera was found to be significantly altered in the KP group, and the LEfSe results suggested the abundance of genera belonging to periodontitis pathogens, namely *Streptococcus*, *Prevotella*, and *Selenomonas* in the KP group. The genera *Lactobacillus*, *Selenemonas*, and *Eubacterium* belonging to the phylum Firmicutes which were highest in KP and WWP groups are related to caries and periodontal disease. The proportion of *Veillonella* is comparatively dominating in the JCP group, and almost resemble with KP group, whereas it is almost negligible in the WWP and control group. The abundance of *Streptococcus* is significantly higher in control group than in other three groups. *Actionmyces* and *Haemophilus* are a periodontal participator; although it was not identified as the predominant, its abundance was found to be the slight in the control and other three groups. The organization of salivary microbiota into dental plaque biofilms is important for survival of these microorganisms in the human oral cavity. The following early colonizers *Streptococcus*, *Actinomyces*, and *Veillonellae* spp. are responsible for plaque biofilm. Additionally, *Haemophilus* are also prominent initiators of cell interactions in the early biofilm formation in oral cavity.

The ANOVA results indicated 14 significantly shifted genera, namely *Capnocytophaga* ($p=0.031$), *Eikenella* ($p=0.037$), *Gemella* ($p=0.008$), *Johnsonella* ($p=0.025$), and *Treponema_2* ($p=0.014$) in the WWC group; *Neisseria* ($p=0.009$) in the WWP group; *Peptostreptococcus* ($p=0.034$) and *Solobacterium* ($p=0.025$) in the JCP group; and *Anaeroglobus* ($p<0.001$), *Howardella* ($p=0.009$), *Prevotella* ($p=0.011$), *Selenomonas* ($p<0.001$), *Selenomonas_4* ($p=0.007$), and *Veillonella* ($p=0.041$) in the KP group.

Alpha Diversity of Salivary Bacterial Communities

The rarefaction curves were found to have attained a stable status, which indicated that the sequencing depth is adequate and can reflect the bacterial diversity of most samples (Fig. 3A). The abundance-based coverage estimators (ACE) index, which can reflect the species richness of a group, were not found to vary significantly between the four groups (Fig. 3B). The Shannon index indicated the highest microbiota diversity in the KP group, and the difference was found to be significant when the microbial diversity of the KP group was compared with that of the WWP group ($p<0.05$) (Fig. 3C). Alpha diversity analysis revealed the highest bacterial diversity in the KP group, and the difference was found to be significant when the bacterial diversity of the KP group was compared with that of the WWP group. These results indicate that the bacterial diversity had increased in patients with periodontitis residing in mining areas. The patients with metal exposure, periodontitis, and SRP (scaling and root planning) treatment resulted in an increase with the metal resistance gene and alter the composition of bacterial profile [6]. A previous study also reported that the abundance and diversity of oral bacteria in patients

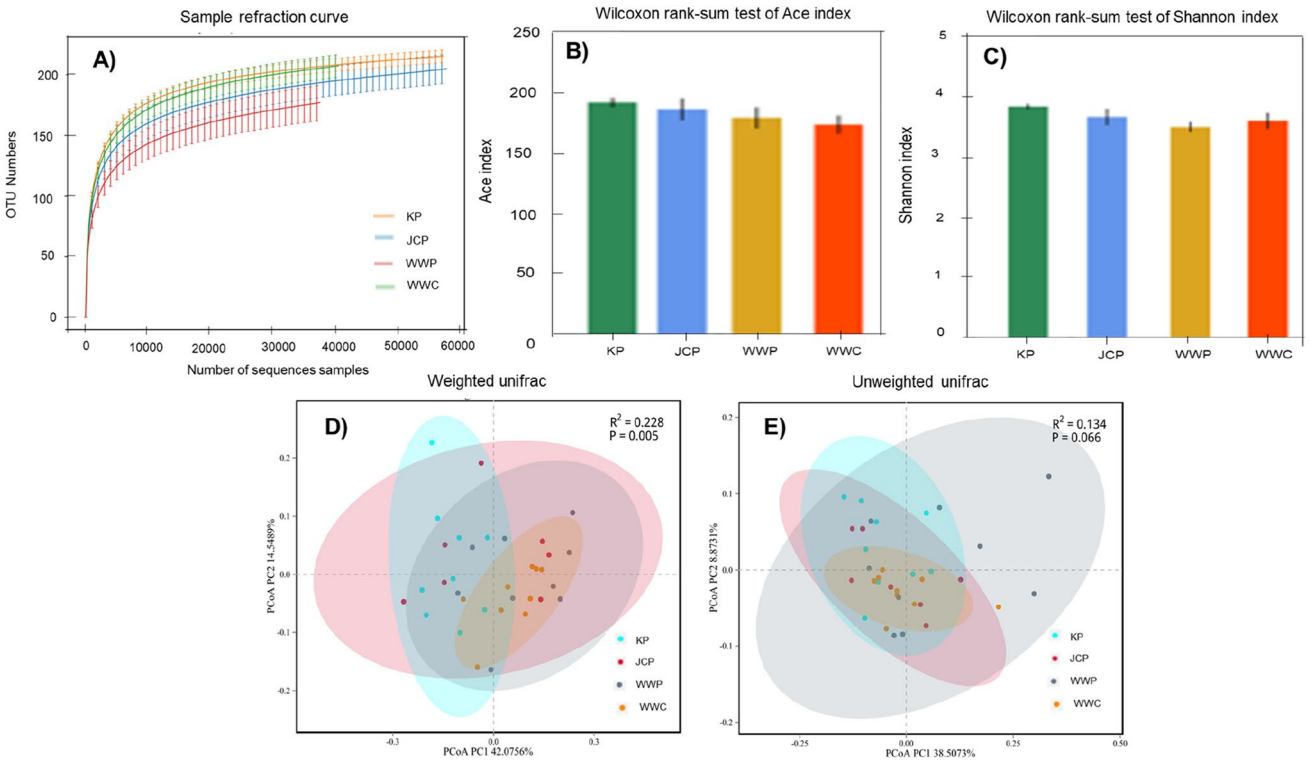


Fig. 3 **A** Sample rarefaction curves. The flat curve indicates that the sample sequence is sufficient for the data analysis; Alpha diversity analysis. **B** Comparison of ACE indices; **C** Comparison of Shannon indices. *Represents a significant difference ($p < 0.05$) in the four groups; principal coordinate analysis (PCoA) derived from weighted (**D**) and unweighted (**E**) UniFrac distances among the samples of the four groups. Every group is represented by different colors

with periodontitis were significantly higher than those in healthy individuals [24].

Beta Diversity of Salivary Bacterial Communities in the Four Groups

To further investigate the difference in OTU abundance between the four groups, the PCoA based on the weighted and unweighted unifrac analysis were conducted. According to the weighted unifrac, the first and second primary components accounted for 42.08% and 14.55%. While in the case of unweighted unifrac analysis, the first and second primary components were 38.51% and 8.87%. (Figs. 3D and 3E). The community population distributions of four groups were found to be overlapped; however, the weighted unifrac-PERMANOVA suggested that the microbiota distribution varied significantly among the groups ($p = 0.005$). These results indicated that despite the comparable composition of the bacterial population, the bacterial abundance varied across the groups. Beta diversity showed the comparable distributions of bacteria in saliva in the four groups, although the difference in abundance was found to be significant between the groups. In agreement with the previous study, beta-diversity analysis consistently showed no differences among the different groups of individuals (healthy and associated with periodontitis) [25]. This finding confirmed that the difference in bacterial abundance may be related to environmental factors and that the local environment determines which bacteria can settle, grow, and establish the dominant population [26].

Analysis of Microbiota Abundance Between the Groups

The LEfSe analysis represents the circles radiating from the inside to the outside of the branch diagram which represent the classification level from phylum to species (Fig. 4). The major abundance in the control groups was *Flavobacteriaceae*, *Capnocytophaga*, *Pasteurellaceae*, *Pasteurellales*, *Proteobacteria*, and *Gammaproteobacteria*, while in the WWP group, the dominating abundance was *Betaproteobacteriales*, *Neisseriaceae*, *Neisseria*, *Flavobacteriales*, and *Flavobacteriaceae*. The main salivary bacterial component in the JCP group was found to be *Cardiobacteriales*. The phylum *Firmicutes*, class *Negativicutes*, family *Veillonellaxeae*, order *Selenomonadales*, and genera *Prevotella* and *Selenomonas* were found to be enriched in the KP group (Fig. 5). Moreover, alterations in the bacterial population of

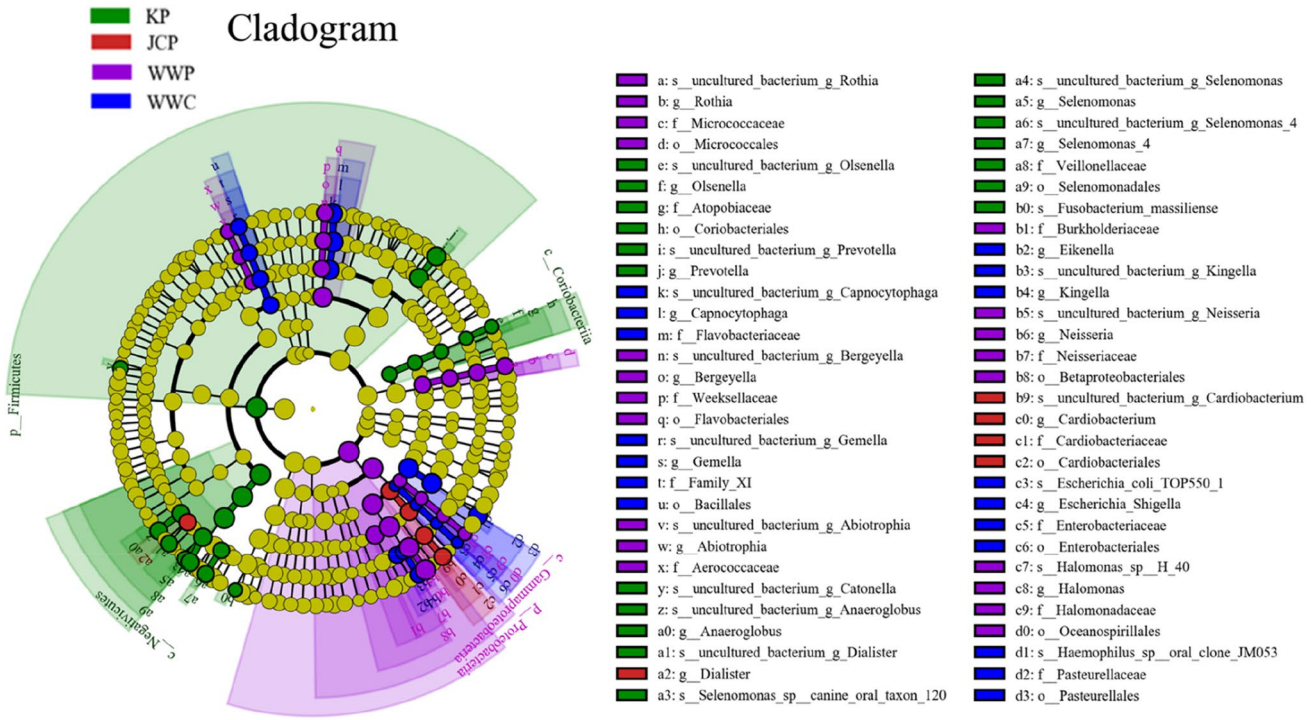


Fig. 4 The LEfSe analysis represents the circles radiating from the inside to the outside of the branch diagram which represent the classification level from phylum to species; each small circle at different classification levels represents a classification at that level, and the diameter of the small circle is proportional to the relative abundance

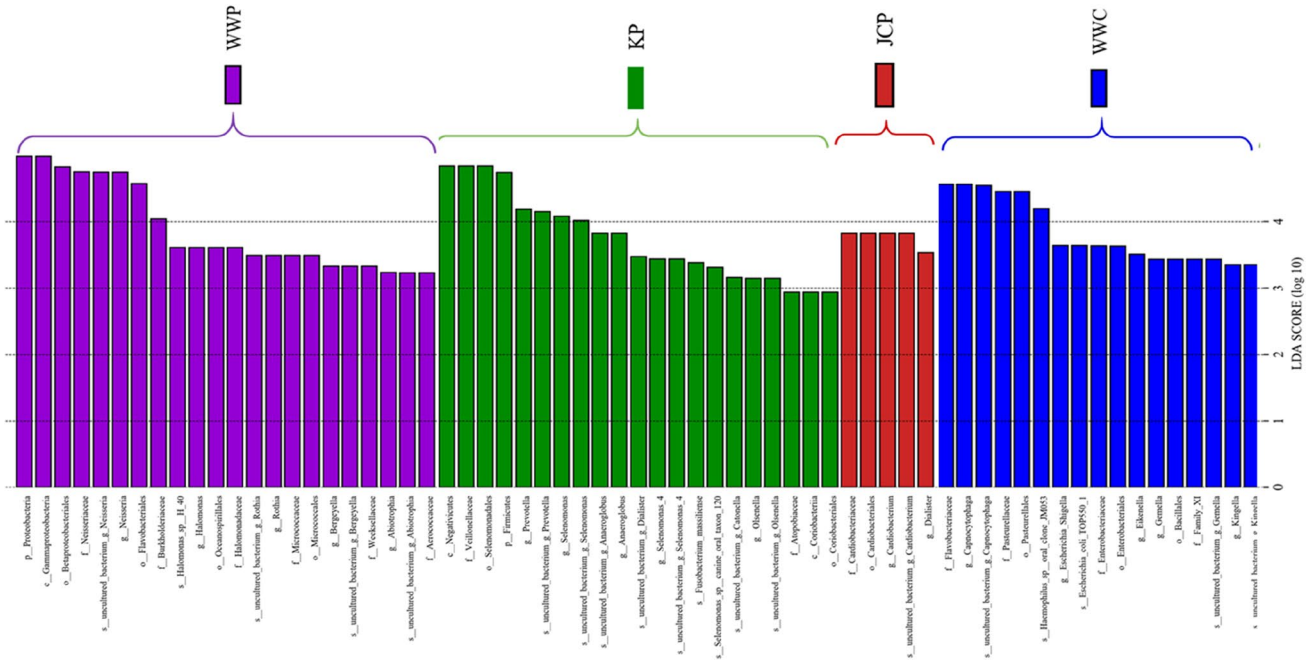


Fig. 5 The significance taxa were tested by LEfSe analysis and showed using histogram. The histogram graphically represents the LDA scores for differentially abundant features among four groups

low abundance often lead to the occurrence and development of periodontal diseases. Some experiments have shown that due to the complex constitution of bacterial populations in the oral cavity, even a small change may disrupt the balance of the bacterial community and result in systemic diseases [27]. A single pathogen is incapable of causing dental caries or periodontal disease, and the cause of periodontal disease appears to be the change in bacterial components originally present in low abundance in the community, resulting in the ecological imbalance [28]. It has been hypothesized that core microbiota is present in the oral cavity, which suggests that primary components are bacterial populations exiting in healthy individuals [13]. The environmental changes cause alterations in the bacterial community structure, especially in the genera having an abundance of < 1%. More specifically, the environmental factors may favor the selection or outgrowth of some specific bacteria.

The Correlation Analysis Between Microbiota and Environmental Factors

The content of heavy metals of Jinchang nickel smeltery is significantly higher than that of the other two areas (Table 2). The pH value of the soil collected from Jinchang nickel smeltery, Jinchang Baye hospital, and Wuwei people's hospital was 9.04, 8.09, and 8.2, respectively. The pH value of the soil is alkaline and the coefficient of variation is low. The effects of human activities and movement, plants, terrain, and climate may cause certain errors in the results. A positive correlation of *Capnocytophaga*, *Neisseria*, and *Streptococcus*, were associated with heavy metals and pH whereas a negative correlation of *Treponema_2*, *Alloprevotella*, *Fusobacterium*, *Prevotella_7*, *Prevotella_2*, *Prevotella*, and *Selenomonas_3* were found with heavy metals and pH (Fig. 6). There was no significant association observed within the microbiota of *Lachnoanaerobaculum*, *Leptotrichia*, *Porphyromonas*, *uncultured_bacterium_f_Saccharimonadaceae*, *Veillonella*, *Actinomyces*, and *Corynebacterium* with environmental factors which were mostly found in the WWC and KP groups. *Capnocytophaga*, *Neisseria*, *Aggregatibacter*, and *Streptococcus* were found to be positively correlated with the content of Cr, Ni, and Cu, whereas *Campylobacter*, *Selenomonas*, *Prevotella_2*, *Prevotella*, and *Selenomonas_3* were found to be negatively correlated with the content of these three HMs. The findings of this study indicate that the heavy metals present in the soil (Cr, Ni, and Cu) are associated with the growth of *Capnocytophaga*, *Neisseria*, *Aggregatibacter*, *Streptococcus*, *Campylobacter*, *Selenomonas*, and *Prevotella* and the environmental factors cause changes in the bacterial community structure. Therefore, the increased level of HMs in soils of the Jinchang mining area could promote the development of periodontitis by affecting the abundance and diversity of the oral microbiota.

The environmental factors alter the proportions of human oral microbiota, which causes damage to the microbial interactions, and eventually leads to periodontal disease [29]. Long-term exposure to HMs, such as nickel, cobalt, lead, and chromium was reported to affect the expression of pro-inflammatory mediators and lead to the destruction of periodontal tissue [30]. In a previous study, it is reported that the lead in the blood for a long time can have negative oral health outcomes; the elevated level of lead can increase the expression of IL-32, CD44, NFKBIA, CTSC, TNFAIP3, IL-1A, IL-1B, IL-8, and CCL20 in the dental plaque containing *Streptococcus gordonii*, *Fusobacterium nucleatum*, and *Porphyromonas gingivalis* and subsequently increase the activity of some inflammatory pathways [31]. These inflammatory factors, including immunological suppression and metabolism [32], can promote the

Table 2 The heavy metal (HM) content is present in different soil samples from various regions

Sampling site	Parameters	Nickel (Ni)	Chromium (Cr)	Copper (Cu)
Jinchang nickel smeltery	Maximum	237.4	193.7	206.7
	Minimum	186.7	165.4	185.4
	Average	215.32	174.95	197.32
	Standard deviation	14.29	7.96	6.46
	Coefficient of variation	6.64%	4.55%	3.27%
Jinchang Baye hospital	Maximum	100.3	98.9	100.6
	Minimum	74.8	67.3	72.3
	Average	89.78	82.94	86.26
	Standard deviation	7.99	7.14	7.17
	Coefficient of variation	8.90%	8.61%	8.31%
Wuwei people's hospital	Maximum	32.4	33.9	31.5
	Minimum	20.4	23.3	20.5
	Average	29.03	29.56	25.77
	Standard deviation	2.59	3.14	3.25
	Coefficient of variation	8.92%	10.62%	12.61%

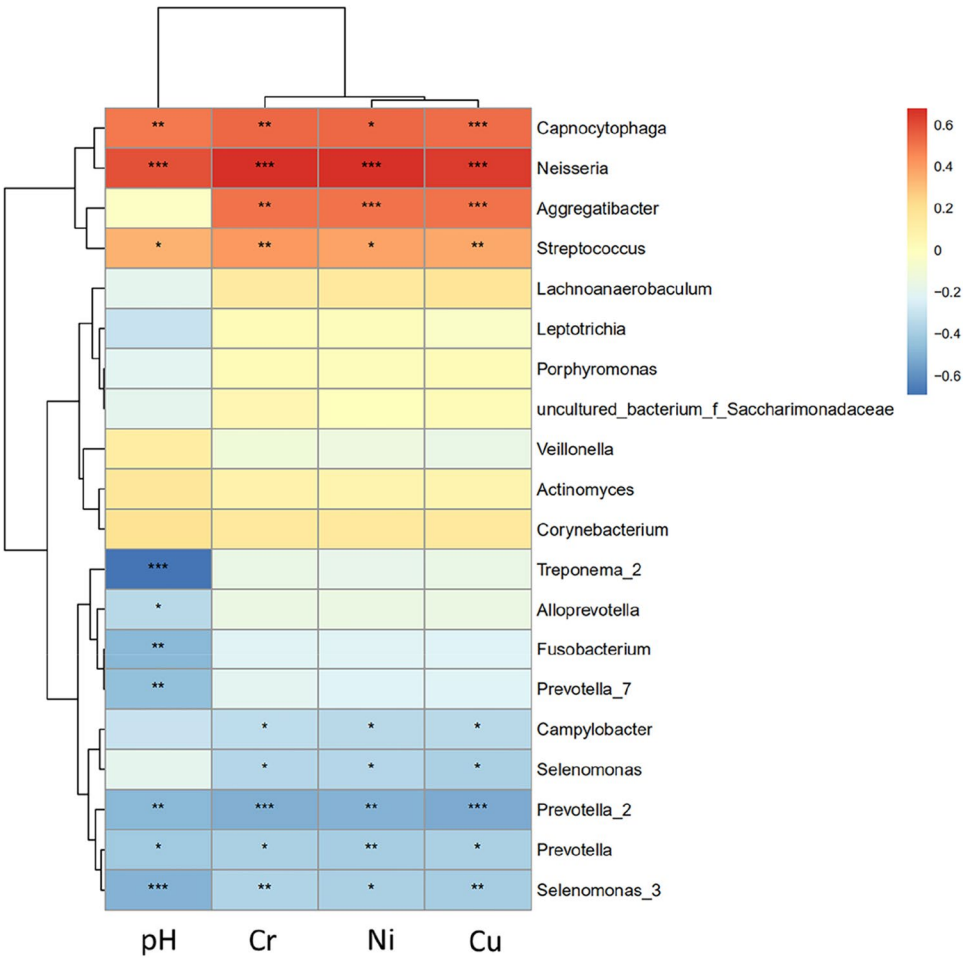


Fig. 6 The correlation between the genus and environment factors (*$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$; p value was calculated using the SparCC algorithm). Copper (Cu), nickel (Ni), chromium (Cr), and pH indicating the correlation with salivary microbiota

occurrence and development of periodontitis. According to the Third National Health and Nutrition Examination Survey (NHANES III) (1988–1994) data, the elevated level of lead in the blood significantly increases the risk of periodontitis, regardless of gender [33]. In a study, the pH value was reported to influence the solubility and bioavailability of heavy metals [34]. Even at a low concentration, the HMs in saliva can impair oral health and change the bacterial community structure.

The number of bacteria and biofilm density was higher on the Co-Cr alloy than on the titanium alloy, suggesting that the metal corrosion and ion precipitation can cause a change in the bacterial community structure [35]. Arsenic, a common HM pollutant, was reported to change the intestinal microbiota of mice and result in a significant decrease in the abundance of phylum Firmicutes [36]. Moreover, a shift in the intestinal bacterial community was observed in mice upon exposure to As and Cr, with a higher concentration of the phylum Bacteroides observed in the exposed mice than that in the control mice [37]. Although there are still limited studies to support the literature of the impact of HM exposure on the oral microbiome community structure, alterations in the intestinal microbiota due to exposure to heavy metals are incontrovertible; meanwhile, the homology between oral and intestinal microbiota is well known [38]. The HM pollution of soils inevitably alters the microbiome community structure of soils and subsequently affects the human body through the food chain [39].

The Predictive Function Analysis Using KEGG

The differences in microbial gene functions in terms of metabolic pathways between the four groups were evaluated using the KEGG enrichment analysis. Overall, the difference was found to be significant between the KP and WWC groups (Fig. 7A) and between the KP and WWP groups (Fig. 7B). However, no significant difference was observed between the WWP and WWC groups and between the KP and JCP groups. The level 2 KEGG pathway indicated that the microbial gene functions, namely lipid metabolism, amino acid metabolism, transport and catabolism, immune diseases, infectious diseases, and parasitic and circulatory system, were lower in the KP group than in the WWC group ($p < 0.05$). In addition, microbial gene functions in the KP

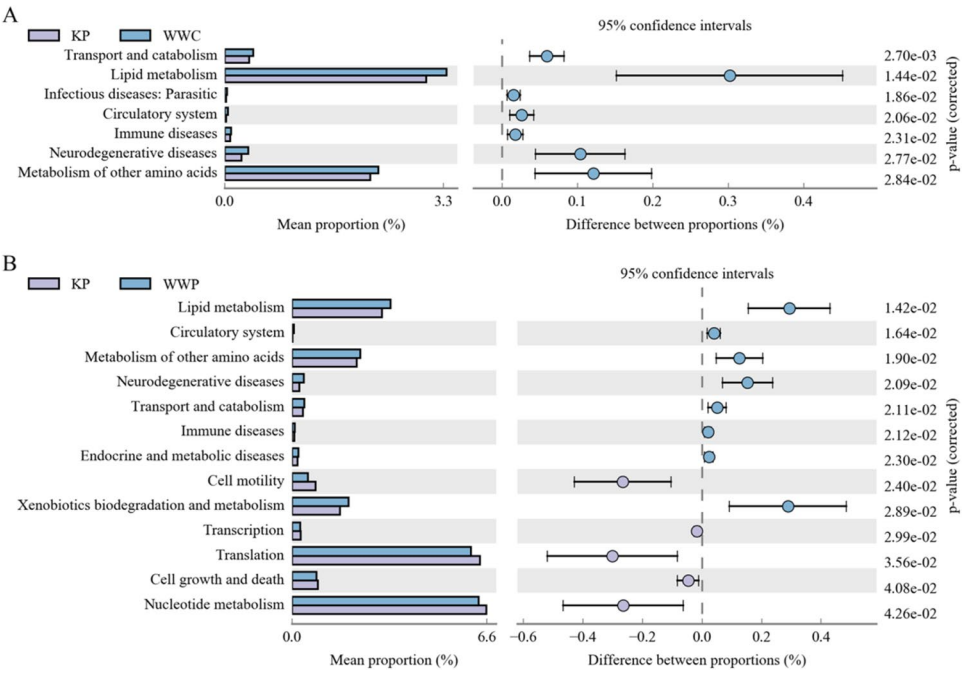


Fig. 7 The difference analysis diagram of the KEGG metabolic pathway at the second level. The middle shows the difference ratio of the function abundance within the 95% confidence interval, and the rightmost value is the *p* value. Analysis between WWC and KP groups (**A**) and WWP and KP groups (**B**)

group were found to be lower in immune diseases, endocrine and metabolic diseases, cell motility, xenobiotics biodegradation and metabolism, and transcription ($p < 0.05$), however higher in translation, cell growth, and death and nucleotide metabolism compared with the WWP group ($p < 0.05$). The microbiota exists in the oral cavity in the form of a biomembrane or biofilm, and its constitution and metabolic activity depend on the host and the environment [40].

The present study revealed no significant differences in the microbial gene functions between the WWC and WWP groups, as well as between the JCP and KP groups. In contrast, the KP group displayed significantly different microbial gene functions compared with the WWC and WWP groups. These changes in the salivary microbiota and metabolic functions are suggestive of tolerance and stress response of bacteria to the environment. The soil HMs can enter human bodies directly or indirectly through the food chain and subsequently impact the constitution and growth of bacterial populations on saliva. The results suggested that higher bacterial metabolism functions in the periodontitis group than in the periodontally healthy group. The finding reflected that the development of periodontitis somehow inhibited the formation of secondary metabolites and protein biosynthesis in bacteria. The underlying mechanism may be the decreased nutrient availability to oral bacteria, resulting in the production of harmful materials by bacteria, which eventually leads to the development of periodontitis [41]. The HMs can accelerate the development of periodontitis through various mechanisms such as disruption of cell mebrane integrity, inhibition of the enzyme activity, and inflammation [42]. Similarly, the presence of copper ions can inhibit the adhesion of periodontal pathogens (*Prevotella intermedia* and *Prevotella nigra*) to the epithelial cells [43]. A brief summary of the results of the current study in comparison with the other previously reported studies is shown in Table 3. Furthermore, it is well known that HM ions may also participate in gene transcription, translation, nucleotide metabolism, and other aspects of cell activities, which accelerates the cell cycle of bacterial cells and helps improve the adaptability of bacteria in response to changes in the oral microenvironment caused by heavy metal ions and the stability of oral microbiota, eventually increasing microbiota diversity in patients with periodontitis residing in the mining regions.

Conclusions

Long-term exposure of HMs presents in the soil (Cr, Ni, and Cu) is associated with the growth of *Capnocytophaga*, *Neisseria*, *Aggregatibacter*, *Streptococcus*, *Campylobacter*, and *Selenomonas*. There were significant divergences in the salivary microbial communities among different groups. Each group has its specific high abundance of bacteria. In alpha diversity, the commendatory bacterial diversity was found in the KP group, and a significant difference was present, while in the case of beta diversity, the comparable distributions of abundance were found to be significant between the groups. The content of heavy metals in the environment is not limited to the soil; hence, the future studies can be focused on water, air, and other environmental factors. This study will provide additional insights into the oral health

Table 3 Summary of the environmental factors, major dominating bacterial community, and the overall outcome of this study compared with the previously reported studies

Research object	Environmental factors	Dominating microbial community	Study outcome	References
Early stage Parkinson's disease (PD)	ND	Bifidobacteriacea, Lactobacillaceae	The oral microbiome represented a highly accessible and informative microenvironment that offers new insights into the pathophysiology of early stage PD	[43]
Rheumatoid arthritis (RA)	ND	*Selenomonas noxia*, *Parvimonas micra*, *Fusobacterium nucleatum*, *Treponema socransky*	Periodontitis was associated with increased RA severity and treatment of periodontitis improved the symptoms of RA	[44]
IgA nephropathy vs chronic periodontitis (CP)	ND	Proteobacteria, Actinobacteria, Fusobacteria, Spirochaetae, Synergistetes	Changes in subgingival microbial structure were used to predict the IgAN incidence in CP patients	[14]
Periodontitis	Heavy metals	ND	Long term exposure affects the levels of the pro-inflammatory mediators leading to more periodontal tissue destruction	[45]
Healthy children, adults, caries	Heavy metals	*Lactobacillus_sp_11808*, *Prevotella pleuritidis*, *and Escherichia coli*	This study hypothesized that the levels of salivary nutrients and metals correlate with the salivary microbiome composition associated with dental decay	[46]
Soils	Metals	Acidobacteria_Gp and Proteobacteria_thiobacillus	The archaeal community were resistant to heavy metal contamination and contributed to the adaption of heavy metals	[47]
Periodontitis	Soil and heavy metals	*Leptotrichia*, *Neisseria*, *Fusobacterium*, and *Capnocytophaga*	This study provided additional insights into the oral health care and periodontitis disease control measures	This study

ND not determined

care and disease control measures for the population residing in the Jinchang mining area.

Author Contribution Wei Zhang: methodology, visualization, investigation, data curation, formal analysis, writing, original draft. Tao Qi: investigation, data curation, formal analysis, writing — review and editing. Lihe Yao: visualization, formal analysis. Wei Wang: visualization, formal analysis. Fanrong Yu: visualization, formal analysis. Yuqin Yan: investigation, data curation, formal analysis. El-Sayed Salama: data curation, validation, visualization, formal analysis, writing — review and editing. Shaochen Su: conceptualization, supervision, resources, data curation, validation, writing — review and editing, funding acquisition, project administration. Ming Bai: conceptualization, supervision, resources, data curation, validation, writing — review and editing.

Funding This study was financially supported by the National Natural Science Foundation of China (31500412), National Natural Science Foundation of Gansu Province, China (20JR5RA327, 21JR1RA152), and Lanzhou science and technology plan project (2018–4-66).

Data Availability The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found below: https://www.ncbi. nlm.nih.gov/sra/PRJNA721290.

Code Availability Not applicable.

Declarations

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Conflict of Interest The authors declare no competing interests.

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